FFSD PROJECT

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xploreo2324@gmail.com

❖ TRAVEL AND HOSPITALITY

➢ Travel Itinerary and Coordination Platform

- **Team Members** :

1. S20240010012 — A.Rushi
2. S20240010052 — Dheeraj.D
3. S20240010080 — G.Siddarth
4. S20240010119 — K.Sreekar
5. S20240010123 — K.Sriram

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1.0 INTRODUCTION:

This document describes the requirements of a tourism and trip-planning platform. The system is designed to help travelers plan, manage, and customize their trips using a single platform.

The platform connects travelers with local guides and service partners such as transport providers, accommodation services, dining services, and activity organizers. It aims to simplify trip planning, reduce manual effort, and improve coordination among all users.

This document explains the system's purpose, scope, functional and non-functional requirements, activity and sequence diagrams. It helps developers, designers, and other users understand how the system works and what it is meant to do.

2.0 VERSION HISTORY:

Version	Date	Description
1.0	Jan 2026	Initial development of the project and requirement gathering

(Further updates are to be added when a new version is made)

3.0 REQUIREMENTS:

3.1 FUNCTIONAL REQUIREMENTS:

1. User Account Management

- Allow users to register on the platform.
- Enable secure login and logout functionality.
- Allow users to view and update their personal profile details.
- Maintain accurate and up-to-date user information.

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2. Travel Search and Booking

- Allow users to search for travel services such as flights, trains, buses, and hotels.
- Display relevant travel options based on user inputs.
- Enable users to book selected travel services through a single interface.

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3. Booking Modification and Cancellation

- Allow users to modify existing bookings when required.
- Enable users to cancel confirmed bookings.
- Provide flexibility to manage changes in travel plans.

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4. Trip Cost Calculation and Display

- Calculate the total trip cost based on selected services.
- Include taxes, discounts, and applicable offers in the cost.
- Display the final payable amount clearly before booking confirmation.

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5. Booking Notifications and Updates

- Send booking confirmation messages to users.
- Notify users about booking updates or changes.
- Deliver notifications through email or SMS for timely communication.

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6. Personalized Travel Plan Recommendations

- Collect user preferences such as budget, interests, and travel requirements.
- Generate travel plan suggestions based on collected preferences.
- Assist users in making informed travel decisions.

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7. Preference Update and Plan Re-suggestion

- Allow users to update their travel preferences at any stage.
- Re-generate travel plan suggestions based on updated preferences.
- Ensure flexibility in trip planning and customization.

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8. End-to-End Itinerary Visualization

- Display the complete trip itinerary in a structured format.
- Show travel and accommodation details for each stage of the journey.

- Enable users to view the trip from start location to final destination.

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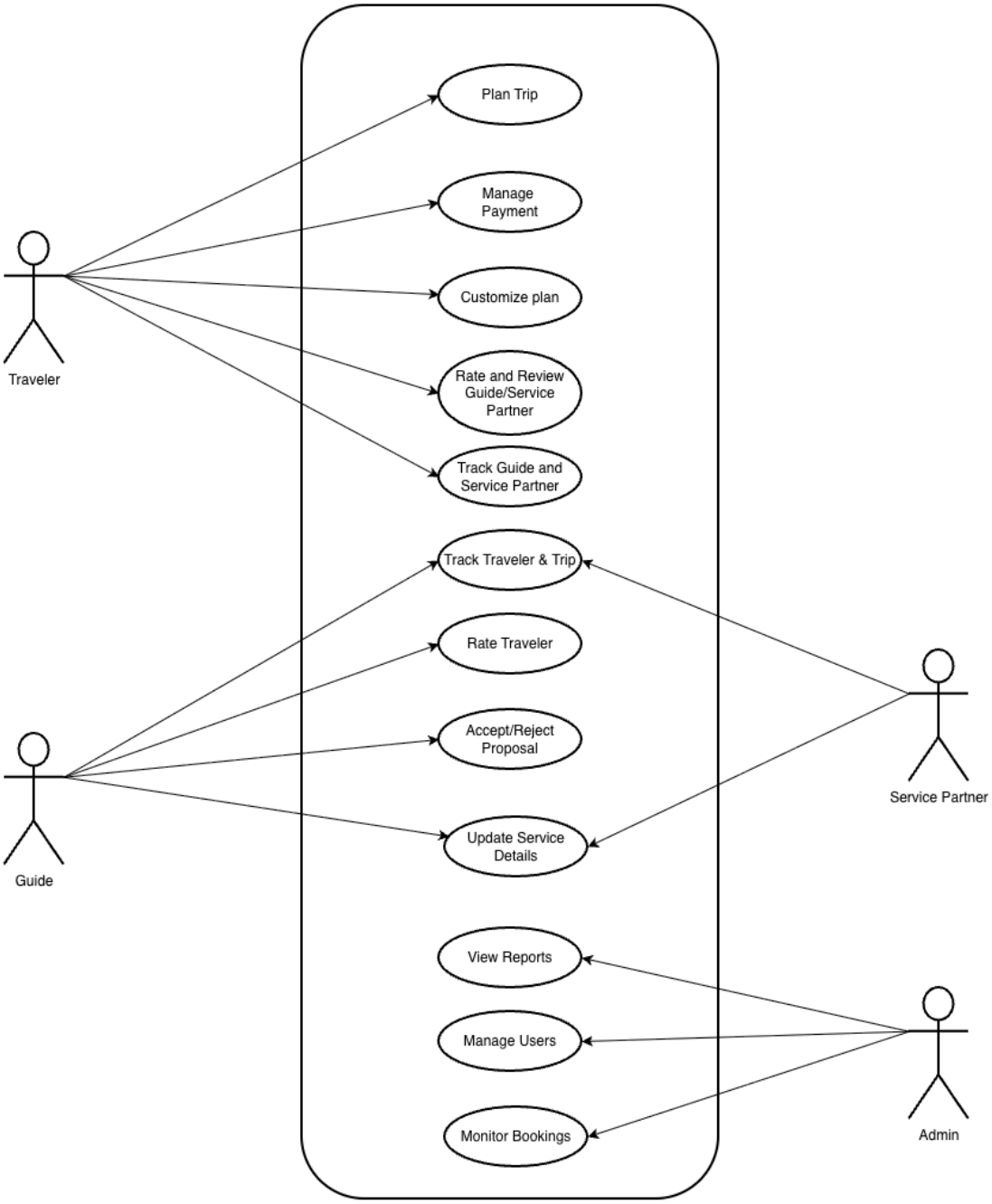
9. Trip Status and Progress Tracking

- Track the progress of the trip throughout the journey.
- Categorize trip stages as completed, ongoing, or upcoming.
- Provide users with real-time awareness of their journey status.

3.2 NON-FUNCTIONAL REQUIREMENTS:

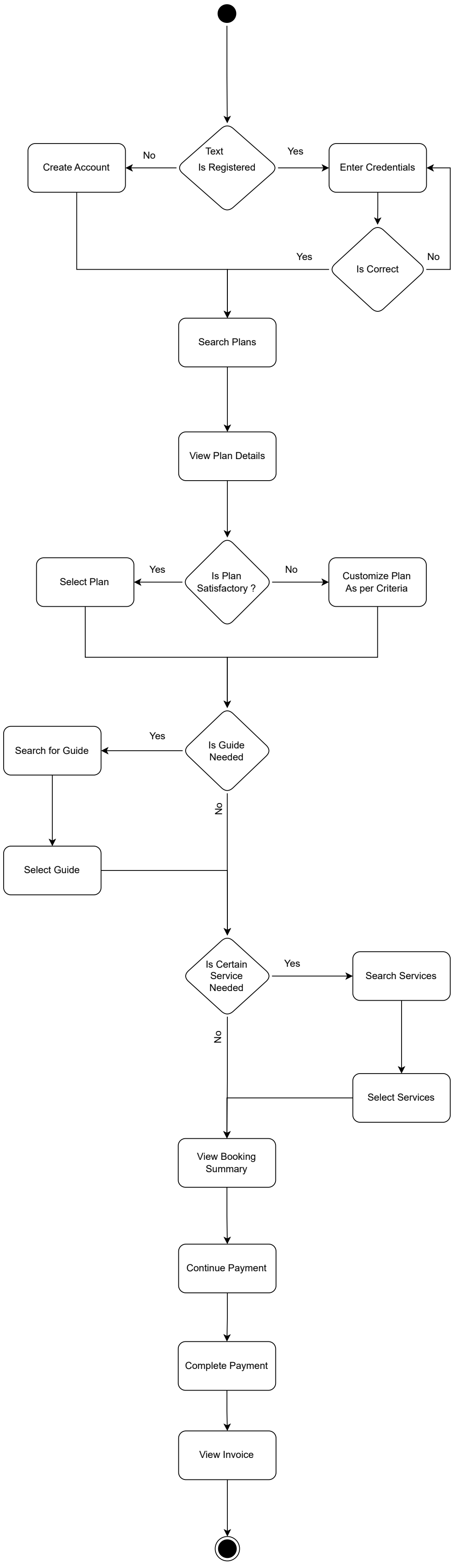
Performance	Search results should be displayed within 3 seconds, and travel package recommendations generated by the system should be available within 10–15 seconds, depending on user preferences.
Scalability	The system should be able to handle high user traffic, especially during peak travel seasons.
Security	User data should be securely stored using encryption, and the system should provide safe and reliable user authentication.
Availability	The system should be available 24/7 with at least 98% uptime.
Usability	The user interface should be simple, clear, and easy to use, even for first-time or non-technical users.

4.0 USE-CASE DIAGRAM:

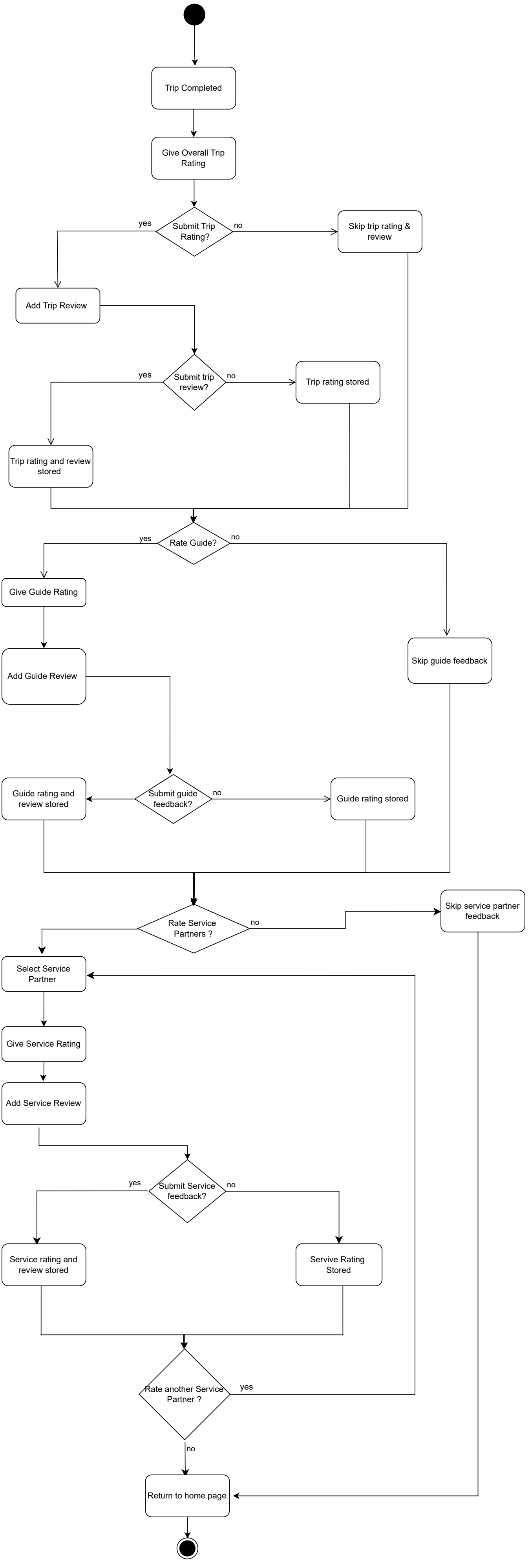


5.0 ACTIVITY DIAGRAMS:

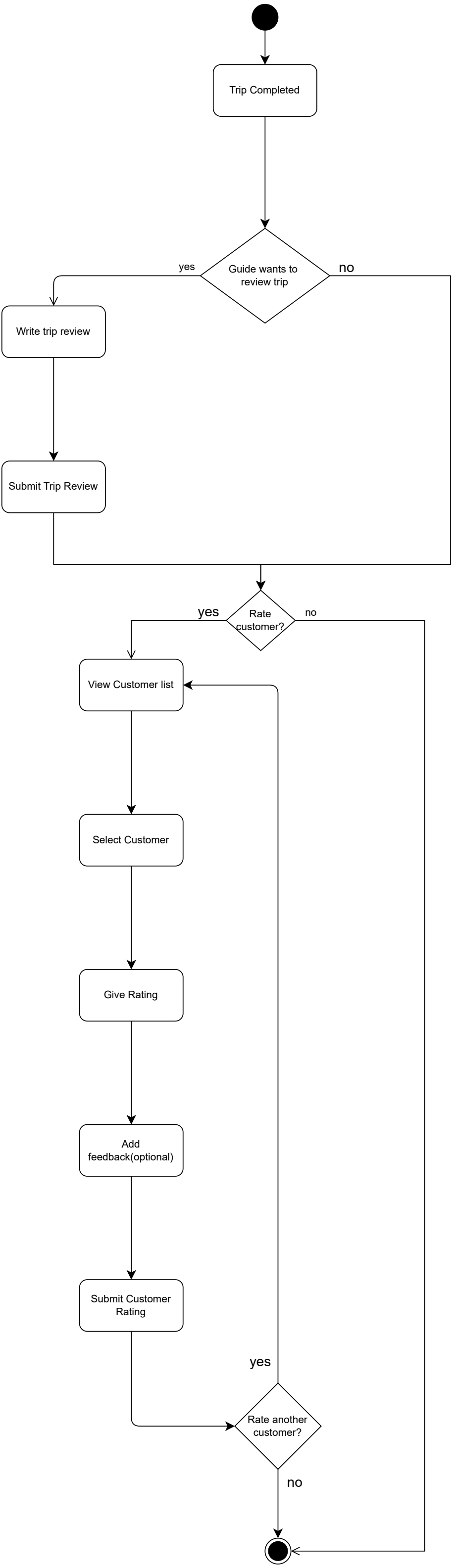
5.1. Plan Trip Activity Diagram:



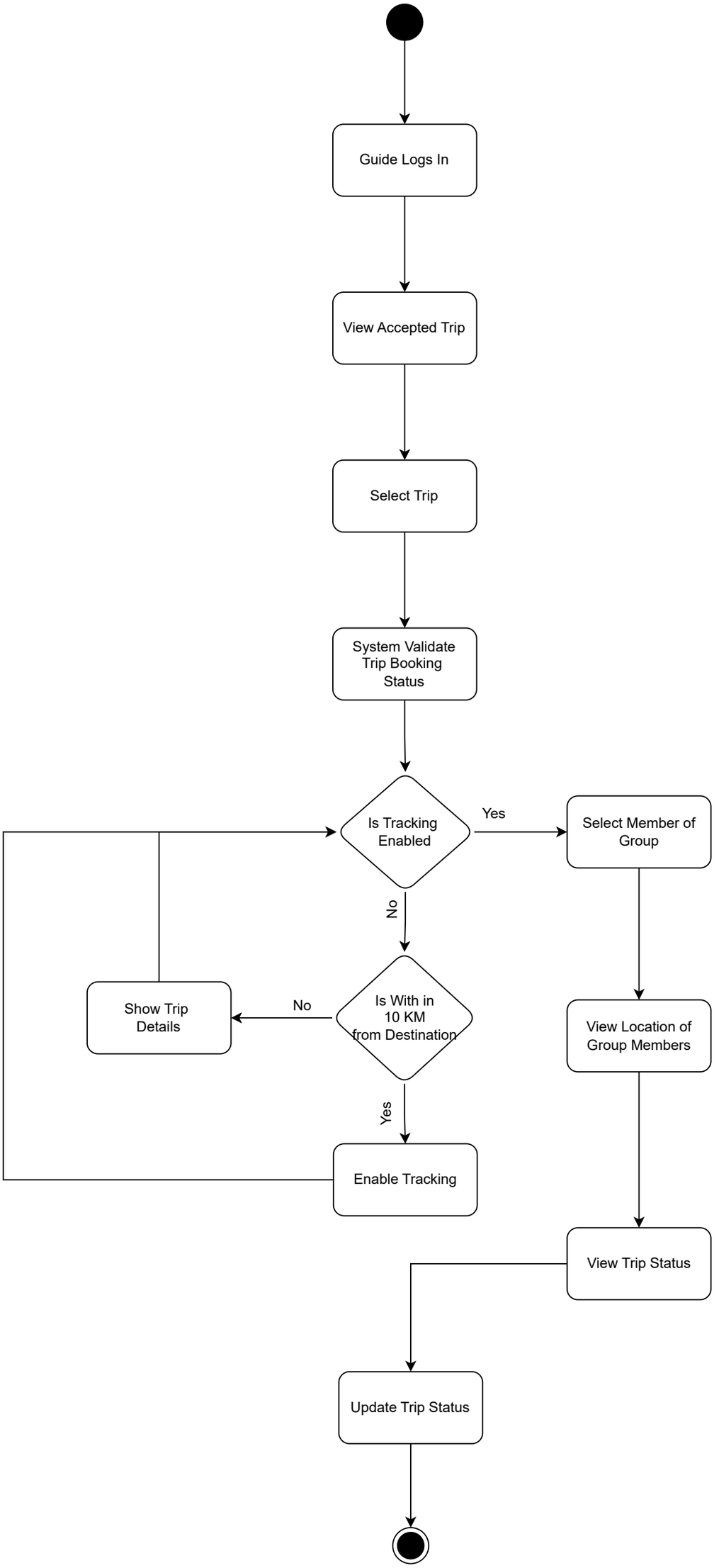
5.2. Ratings and Reviews Activity Diagram (Traveler):



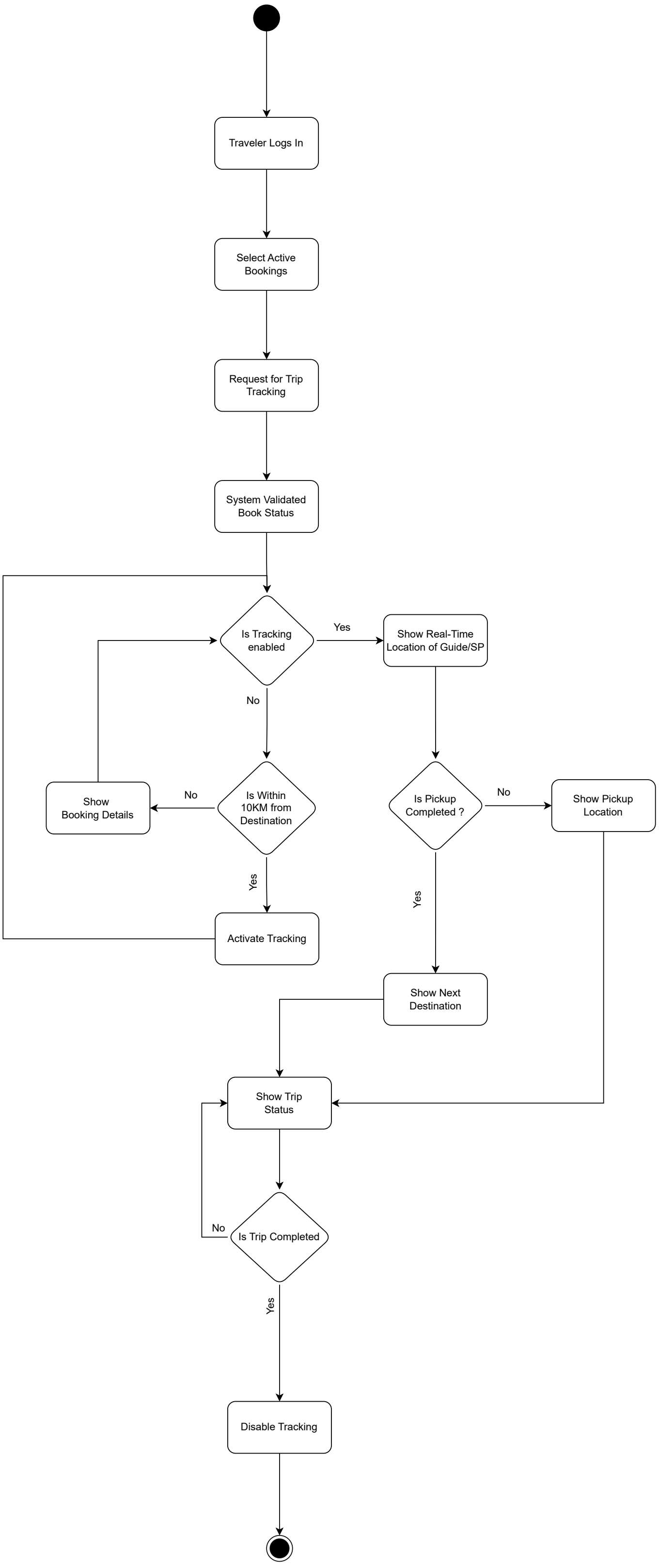
5.3. Ratings and Reviews Activity Diagram (Guide):



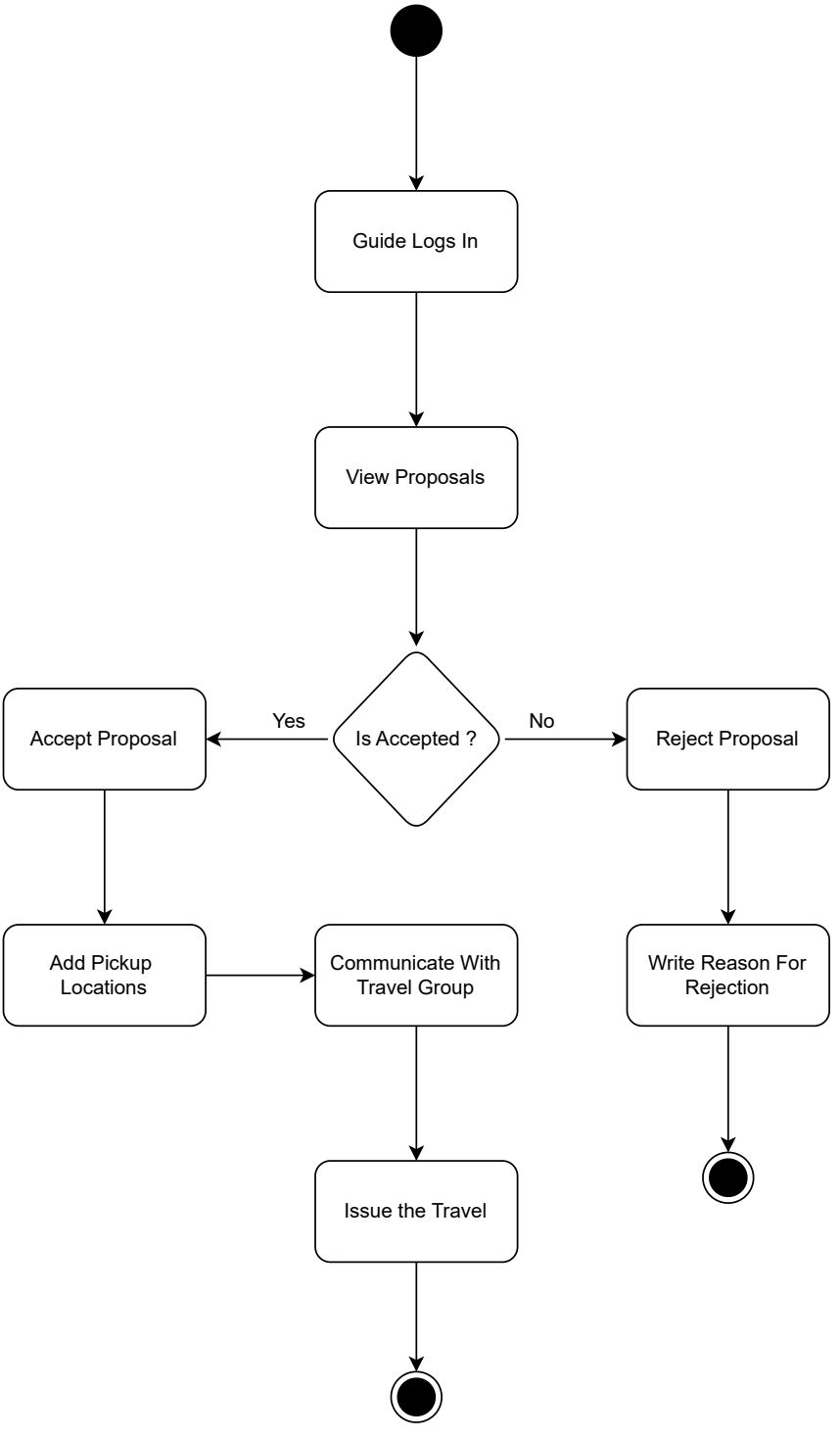
5.4. Tracking Service Activity Diagram (Guide):



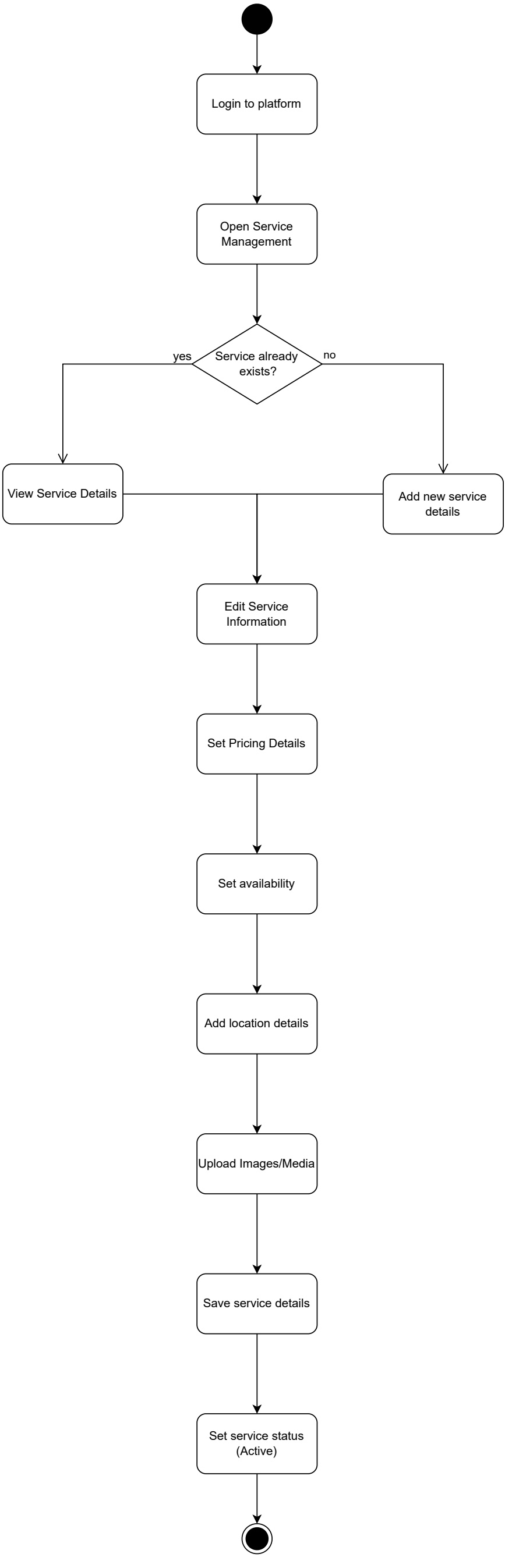
5.5. Tracking Services Activity Diagram (Traveler):



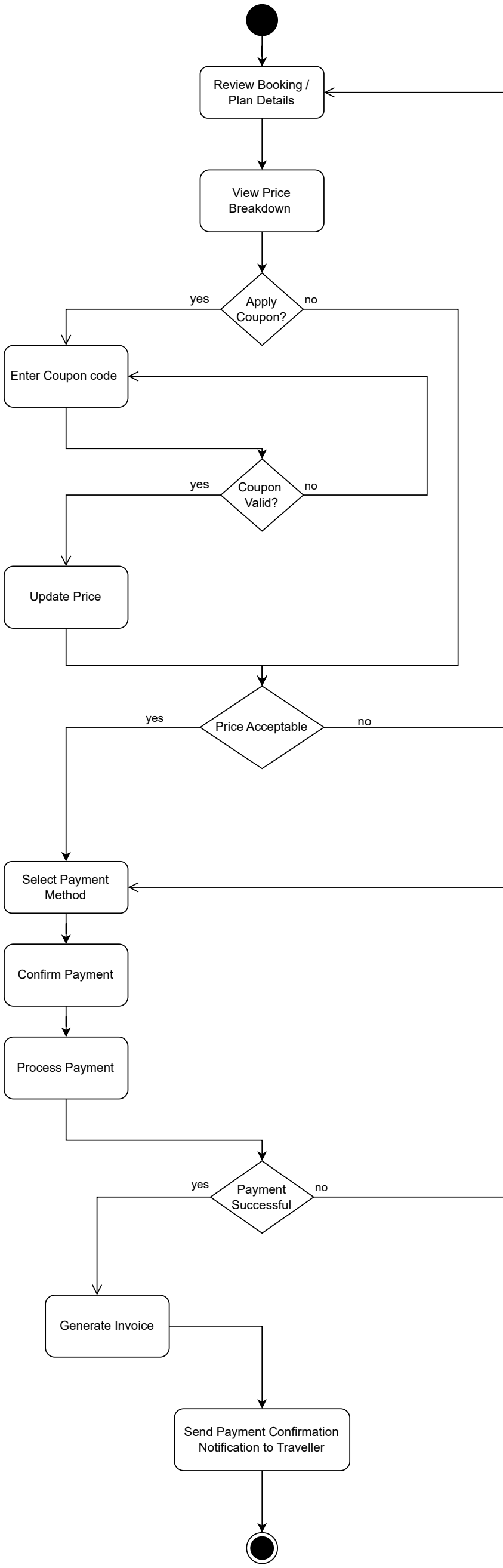
5.6. Manage Proposals Activity Diagram:

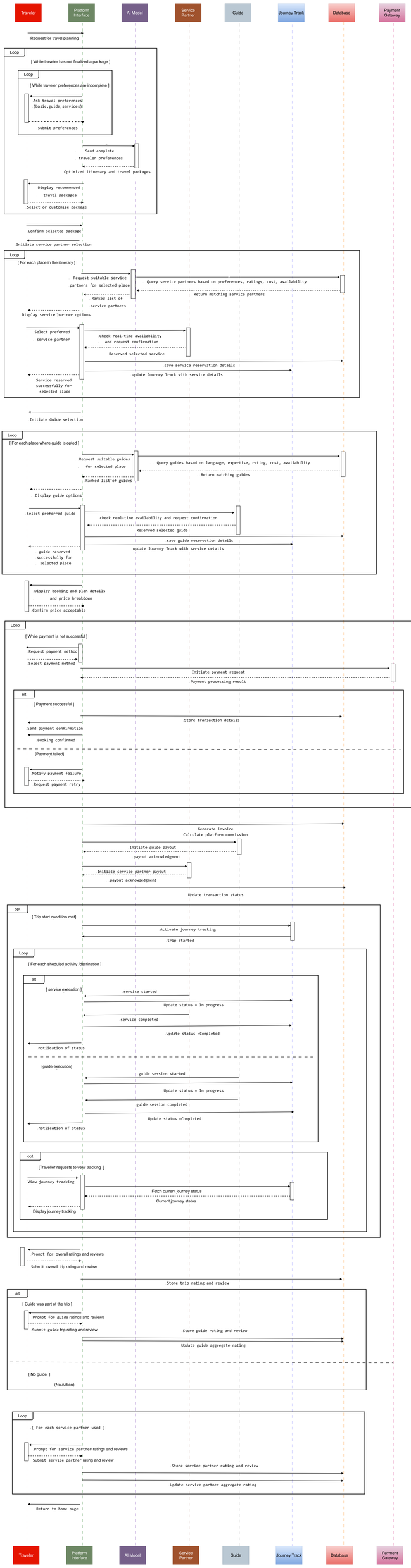


5.7. Service Details Activity Diagram:



5.8. Payment Process Activity Diagram:




Traveler
Platform Interface
AI Model
Service Partner
Guide
Journey Track
Database
Payment Gateway
Request for travel planning
Loop
[While traveler has not finalized a package]
Loop
[While traveler preferences are incomplete]
Ask travel preferences (basic,guide,services)
submit preferences
Send complete traveler preferences
Optimized itinerary and travel packages
Display recommended travel packages
Select or customize package
Confirm selected package
Initiate service partner selection
Loop
[For each place in the itinerary]
Request suitable service partners for selected place
Query service partners based on preferences, ratings, cost, availability
Return matching service partners
Ranked list of service partners
Display service partner options
Select preferred service partner
Check real-time availability and request confirmation
Reserved selected service
save service reservation details
update Journey Track with service details
Service reserved successfully for selected place
Initiate Guide selection
Loop
[For each place where guide is opted]
Request suitable guides for selected place
Query guides based on language, expertise, rating, cost, availability
Return matching guides
Ranked list of guides
Display guide options
Select preferred guide
check real-time availability and request confirmation
Reserved selected guide
save guide reservation details
update Journey Track with service details
guide reserved successfully for selected place
Display booking and plan details and price breakdown
Confirm price acceptable
Loop
[While payment is not successful]
Request payment method
Select payment method
Initiate payment request
Payment processing result
alt
[Payment successful]
Store transaction details
Send payment confirmation
Booking confirmed
[Payment failed]
Notify payment failure
Request payment retry
Generate invoice
Calculate platform commission
Initiate guide payout
payout acknowledgment
Initiate service partner payout
payout acknowledgment
Update transaction status
opt
[Trip start condition met]
Activate journey tracking
trip started
Loop
[For each sheduled activity /destination]
alt
[service execution]
service started
Update status = In progress
service completed
Update status =Completed
notiication of status
[guide execution]
guide session started
Update status = In progress
guide session completed
Update status =Completed
notiication of status
opt
[Traveller requests to view tracking]
View journey tracking
Fetch current journey status
Current journey status
Display journey tracking
Prompt for overall ratings and reviews
Submit overall trip rating and review
Store trip rating and review
alt
[Guide was part of the trip]
Prompt for guide ratings and reviews
Submit guide trip rating and review
Store guide rating and review
Update guide aggregate rating
[No guide]
(No Action)
Loop
[For each service partner used]
Prompt for service partner ratings and reviews
Submit service partner rating and review
Store service partner rating and review
Update service partner aggregate rating
Return to home page